PROJECT CANARY · TEAM STUDY GUIDE

Capstone Defense Cheat Sheet

What Canary is, how each component works, what the results mean, and which claims the evidence supports

Core story: Identify buildings that need attention, project Day 35 weight and harvest recovery, explain the warning signs, and recommend the next management check.

Updated 11 August 2026. Use this as the shared answer key for the dashboard, notebooks, model files, and presentation.

One-page “memorize this” summary

What is Project Canary?

Project Canary is an early-warning and decision-support system for a broiler farm. It identifies buildings that need attention, explains the recorded warning signs, estimates harvest recovery and Day 35 average weight, and links confirmed operating patterns to Doc Raymond’s action playbook.

What business question does it answer?

Which buildings need attention today, why, what harvest recovery and Day 35 weight should we expect, and what should management check next?

What are the three engines?

1. **Risk scoring:** transparent rules score four observed dimensions from 0–3 points each. Total 0–12 becomes Low, Medium, High, or Critical.
2. **Predictive outlooks:** a recovery estimator and a Day 35 weight method are evaluated independently using data available by the selected review date.
3. **Recommendation playbook:** deterministic triggers map to Doc Raymond’s recorded checks, urgency, and escalation. It does not diagnose disease or automatically treat birds.

What data do we have?

- 1,785 source rows became **1,666 unique building-day records** after repeated Zone A/B environment rows were consolidated.
- Recovery: **25 independent building-cycle outcomes**, represented by **122 balanced decision snapshots**; 1,122 leakage-safe daily snapshots are retained for audit.
- Day 35 weight: **31 independent building-cycle outcomes**, represented by **124 Day 7/14/21/28 checkpoint rows**.
- Recovery endpoint is the agreed proxy: last-recorded population ÷ beginning population. The redesigned **training Y** is the additional population loss after each review date.

- Weight endpoint is observed average bodyweight on Day 35. The redesigned **training Y** is the remaining gain from the current checkpoint to Day 35.
- Day 35 goal: **1,800 g**. Recovery goal: **95%**.

What models are operational?

- **Recovery:** the operational method is the **age-band remaining-loss baseline**. Whole-cycle MAE is **1.40 percentage points**; cycle-balanced MAE is **1.50 points**; RMSE is **1.80 points**; R^2 is **0.222**. Robust Huber is the best learned research challenger (**1.24-point MAE, R^2 0.260**) but remains experimental because it fails the 95% classification gate and is less stable under rolling-origin validation.
- **Day 35 weight: historical remaining gain** remains operational. MAE is **178 g**; cycle-balanced MAE is **182 g**; RMSE is **242 g**; R^2 is **0.126**; **65.3%** are within 200 g. Every learned challenger performs worse on unseen cycles.

What is the honest claim?

Canary is a **validated prototype**, not a production guarantee. Recovery is useful directionally as a continuous estimate. Day 35 weight uses the more defensible transparent fallback because the learned models do not yet improve enough on unseen cycles.

Five-step explanation: Engine 1 — Risk scoring

1. **Question:** Which building shows the most operational concern today?
2. **Inputs:** weight gap versus the approved age target; survival/population-loss evidence; recent mortality; and environmental evidence against age-specific temperature/humidity bands. Missing or stale evidence is disclosed.
3. **Process:** each dimension receives 0–3 points using configurable thresholds. Available points are added; the trace shows each observation, rule, score, and missing dimension.
4. **Output:** total score, Low/Medium/High/Critical label, leading problem pattern, and “why now” evidence.
5. **Limitation:** the score is an operational-priority rule, not a probability of missing 95% recovery or 1,800 g. Environmental thresholds remain subject to farm approval.

Why rules rather than a risk classifier?

- Only 25–31 independent outcomes are available.
- The owner needs to see the exact rule and measurement behind every flag.
- Rules keep operating concern separate from statistical forecasts.
- Thresholds can be reviewed and changed without retraining a model.

Five-step explanation: Engine 2A — Recovery forecast

1. **Question:** Using only what is known today, what final recovery proxy should we expect?
2. **Inputs (X):** age and remaining days; current survival and cumulative loss; mortality level/acceleration; latest weight gap/freshness; temperature/humidity exposure and freshness. Feed is withheld until its unit is confirmed.
3. **Process:** create Day 7/14/21/28/latest snapshots; give every building-cycle equal total weight; impute, scale, tune, and compare models inside nested whole-cycle folds.

4. **Output:** current survival minus predicted remaining loss, plus target gap, empirical range, model version, forecast date, and predictive context.
5. **Limitation:** Y is a last-recorded recovery proxy, only five historical cycles are available, and the model cannot yet distinguish 95% hits from misses better than the majority rule.

Recovery workflow — explain it from start to finish

Stage	What Canary does
Data	Starts from the corrected farm workbook, consolidates Zone A/B rows, and produces one building-day record.
Y output	Calculates remaining loss after the snapshot: current percentage alive − completed-cycle recovery proxy. Final forecast = current survival − predicted remaining loss.
X inputs	Uses only evidence known on the review date: age, current survival, mortality signals, latest weight gap/freshness, and environmental deviations/freshness.
Feature engineering	Converts raw readings into per-bird rates, gaps from approved targets, rolling mortality signals, days outside environmental bands, and staleness flags.
Pre-processing	Inside each training fold only: median-impute missing numeric values, add missingness indicators, and standardize inputs for linear models.
Validation	Removes one complete harvest cycle at a time; the inner loop selects settings using only the remaining cycles. Repeated snapshots are weighted so each building-cycle has equal total influence.
Selection	Compares five declared methods on cycle-balanced MAE, then checks R^2, cycle stability, target-side performance, and champion gates.
Live output	Returns current survival minus expected remaining loss, a likely range, gap to 95%, model version, and non-causal driver context.

Recovery Y target

additional loss Y = current percentage alive − final recovery proxy

predicted final recovery = current percentage alive − predicted additional loss

The value is used only after prediction for evaluation. Current survival may be an X because it is already known on the review date and logically constrains what final recovery can be. Future ending population is never an X.

Recovery model comparison — primary prospective test

Candidate	MAE (pts)	Cycle-balanced MAE (pts)	RMSE (pts)	R^2	Decision
Age-band remaining-loss baseline	**1.40**	**1.50**	**1.80**	**0.222**	**Operational fallback**
Linear remaining-loss regression	1.29	1.40	1.80	0.224	Compared
Ridge remaining-loss regression	1.31	1.43	1.82	0.207	Compared
Robust Huber remaining-loss regression	1.24	1.33	1.76	0.260	Best research challenger
Gradient Boosting remaining-loss	1.25	1.37	1.78	0.243	Compared

Why not automatically deploy Huber because its MAE is best?

Huber improves cycle-balanced MAE by **11.2%** and keeps positive R^2, but its rolling-origin cycle MAE is **1.73 points** versus **1.35** for the age-band baseline, and it does not beat the majority rule for identifying the 95% target side. The strict gate therefore keeps the transparent baseline operational while Huber remains the research champion.

Recovery classification warning

- Target-side accuracy: **86.1%** for the operational baseline.
- Majority baseline: **84.4%** overall.
- Day 14: **84.0%**, equal to the **84.0%** majority baseline.
- At/above-95% recall: **0%**.

Therefore, do not say “the model accurately classifies whether recovery will hit 95%.” Say: **it estimates the continuous recovery level, but target-side classification is not validated.**

Recovery top held-out drivers

Canary reports held-out permutation importance for the Huber research challenger. These are predictive associations—not proof that changing a factor will cause recovery to improve. The live baseline itself uses only current survival and the age-band historical remaining loss.

Five-step explanation: Engine 2B — Day 35 weight

1. **Question:** Given recorded growth so far, what average building weight should we expect on Day 35 against the 1,800 g goal?
2. **Inputs (X):** latest/checkpoint weights; weighing day; weight-to-target ratio; recent and cumulative average daily gain; earlier checkpoint weights available by that date; current survival; mortality; and environmental exposure/freshness.

3. **Process:** use only observed Day 7/14/21/28 weights; hide future checkpoints; give each building-cycle equal total influence; run nested whole-cycle validation and apply strict replacement gates.
4. **Output:** projected Day 35 weight, gram and percent gap to 1,800 g, range, method, measurement day, and staleness.
5. **Limitation:** only 31 independent Day 35 outcomes exist; 26 are below target and five are at/above target; no learned model clears every gate.

Day 35 weight workflow — explain it from start to finish

Stage	What Canary does
Data	Uses corrected, observed building weights and keeps observed values separate from target-curve interpolation.
Y output	Calculates remaining gain: observed Day 35 weight − current checkpoint weight. Final forecast = current weight + predicted remaining gain.
X inputs	At each Day 7/14/21/28 checkpoint, exposes only weights and operating evidence already recorded by that checkpoint. Later weights remain blank.
Feature engineering	Creates measurement age, target ratio/deficit, average daily gain, available prior checkpoint weights, survival, environmental exposure, missingness, and staleness.
Pre-processing	Performs imputation, missingness handling, scaling, feature filtering, and tuning inside training folds—not before the holdout.
Validation	Removes a complete cycle, trains on the other cycles, predicts the unseen cycle, and repeats. Every building-cycle has equal total influence.
Selection	Compares historical remaining gain, checkpoint linear, Ridge, robust Huber, and constrained Gradient Boosting. A learned model must improve cycle-balanced MAE by 10%, keep positive R^2, remain stable, reach 70% within 200 g, and improve target-side usefulness.
Live output	Because no learned model passes, adds the training-only historical remaining gain for the measurement age to the latest observed building weight.

Day 35 weight Y target

remaining gain Y = observed Day 35 weight − current checkpoint weight

projected Day 35 weight = current checkpoint weight + predicted remaining gain

The approved target curve is a known reference used to calculate age-specific progress. It never replaces a missing observed Y.

Weight model comparison — primary prospective test

Candidate	MAE	Cycle-balanced MAE	RMSE	R^2	Within 200 g	Decision
Historical remaining gain	**178 g**	**182 g**	**242 g**	**0.126**	**65.3%**	**Operational fallback**
Checkpoint linear remaining-gain	216 g	208 g	273 g	-0.118	56.5%	Compared
Ridge remaining-gain	207 g	200 g	264 g	-0.045	56.5%	Compared
Robust Huber remaining-gain	226 g	223 g	276 g	-0.144	51.6%	Compared
Gradient Boosting remaining-gain	203 g	207 g	262 g	-0.026	52.4%	Best nonlinear challenger

How historical remaining gain works

For each checkpoint age in the training cycles:

remaining gain = observed Day 35 weight − observed checkpoint weight

Average those gains using training cycles only, then:

projected Day 35 weight = current observed weight + average historical remaining gain for that age

The held-out cycle is excluded when validating, so its Day 35 result cannot influence its own prediction.

Why not use a learned model as the live method?

All learned challengers have negative whole-cycle R^2 and higher cycle-balanced MAE than the historical remaining-gain baseline. The approved gate requires at least 10% improvement, positive R^2, stability, and at least 70% within 200 g. Using the baseline is therefore the honest, defensible choice.

What drives the live weight projection?

The live formula uses three direct items: latest measured weight, its measurement day, and the average historical remaining gain for that checkpoint. Learned-model importances may be shown as research evidence, but they do **not** drive the live forecast.

Day 14 weight proof

- 31 building outcomes tested.
- Day 14 MAE: **181 g**.
- Day 14 RMSE: **237 g**.
- Within 200 g: **58.1%**.

- At/above-1,800 g recall: **0%**.

Day 14 remains the main early-warning checkpoint because it is early enough to act and its observed weight is used in both target-gap monitoring and the later-growth forecast. It is a useful signal, not proof of causality.

Five-step explanation: Engine 3 — Recommendations

1. **Question:** What should management inspect next, given the confirmed warning pattern?
2. **Inputs:** deterministic risk/alert pattern, severity, recorded evidence, measurement freshness, and approval status.
3. **Process:** match the trigger to Doc Raymond's playbook—for example low bodyweight, high mortality, high/low temperature, high/low humidity, abnormal fluctuation, low feed intake, or poor recovery outlook.
4. **Output:** plain-language check, urgency, escalation condition, and applied rule.
5. **Limitation:** recommendations are inspection/management guidance. They do not diagnose disease, estimate causal treatment effects, or prescribe automatic treatment.

Why nested whole-cycle validation?

- **Outer loop:** remove every row from one harvest cycle; train and tune without it; predict it; repeat for all cycles.
- **Inner loop:** choose hyperparameters using only the remaining training cycles.
- Imputation, scaling, and tuning never see the outer test cycle.
- Repeated snapshots are weighted so one building-cycle has the same total influence as another.

This answers the deployment-like question: **can Canary generalize to a future unseen harvest cycle?**

The colleague's building-cycle LOGO test answers an easier question: can it predict one building when other buildings from the same cycle remain in training? Its higher R^2 is not directly comparable.

What we learned from the teammate model

Incorporated

- A structured and reproducible cleaning pipeline.
- Zone A/B aggregation before feature creation.
- Environmental, growth, missingness, and freshness features.
- Group-aware cross-validation and comparison with boosted trees.
- Explicit model comparison and feature-importance reporting.

Not adopted as the primary Canary test

- Treating repeated daily rows as new independent flock outcomes.
- Leaving out one building while keeping other buildings from the same cycle in training.
- Using interpolated early weights that were created with later observations.
- Selecting features on the full dataset before cross-validation.
- Using a pickle or headline score that cannot be reproduced from the corrected canonical workbook.

Defense answer: We took the useful engineering ideas, but retained stricter whole-cycle validation because Canary must generalize to a future cycle, not merely another building in a cycle it has already seen.

Why no SMOTE or oversampling?

- These are regression problems, while standard SMOTE is primarily for classification.
- The scarce evidence is independent building-cycles, not spreadsheet rows.
- Synthetic rows do not create new cycles and can make uncertainty look falsely small.
- Synthetic poultry trajectories may be biologically implausible.

How to interpret MAE, RMSE, R^2, and bias

- **MAE:** typical absolute miss in business units; primary selection metric.
- **RMSE:** penalizes large misses more strongly.
- **R^2:** fraction of variance explained on held-out data; useful diagnostic, not the sole decision rule.
- **Bias:** whether predictions are high or low on average.
- **Target-side accuracy/recall:** whether a model distinguishes the two sides of 95% or 1,800 g; always compare with the majority baseline.

Low R^2 means much variation remains unexplained. It does not erase all usefulness, but it requires prototype positioning, wider uncertainty, and conservative claims.

Seven EDA questions Canary answers

1. **Coverage:** Which cycles, buildings, days, and measures are complete enough to analyze?
2. **Early growth:** Is Day 14 weight associated with Day 35 weight?
3. **Early recovery signal:** Is Day 14 weight associated with the final recovery proxy?
4. **Environment:** Are recorded temperature and humidity within the provisional age-specific bands, and is the evidence balanced enough to compare outcomes?
5. **Survival paths:** When do buildings begin to drift below the expected survival path?
6. **Model accuracy:** How large are forecast errors overall, by checkpoint, and by held-out cycle?
7. **Target attainment:** How often have completed building-cycles reached 1,800 g, 95% recovery, or both?

Defense answer: EDA is used to test the business story, expose data limitations, inform transparent features, and prevent overclaiming. It does not prove that changing one factor will cause a better outcome.

Common panel questions

"Did you train on daily rows as if they were independent flocks?"

No. The independent outcome is one building-cycle. Repeated snapshots give multiple historical decision points, but they receive equal building-cycle weighting and are always held out together by cycle.

"Did you leak future weights?"

No. At a Day 14 snapshot, Day 21/28/35 inputs are blank. The Day 35 value is used only as Y after prediction. Target-curve interpolation is a known reference, not an observed weight.

“Why is current survival allowed in recovery X?”

It is known today and directly limits what final recovery can be under the agreed formula. Future ending population is never used as X. Raw beginning population is excluded unless the farm later approves a meaningful density feature.

“Can feature importance tell the owner to lower temperature by 3°C?”

No. Importance shows association and predictive reliance. The action table must be triggered by an observed threshold and approved farm guidance, not by a causal claim from observational data.

“Are these models reliable?”

They are useful as validated prototype estimates, not guarantees. The recovery Huber challenger passes the continuous-error gate but not the full deployment gate; Canary therefore keeps the transparent age-band fallback. Weight learned models also fail the replacement gates, so Canary uses historical remaining gain.

“What would most improve the models?”

Collect more completed cycles with verified harvest dates/populations, consistent bodyweight sampling, confirmed feed units, complete environment readings, and more outcomes on both sides of the targets.

Where to show proof in the deliverables

- **Canary Methodology page:** five-model tables, gates, timing and cycle performance, drivers, and limitations.
- **Building View:** current evidence, forecast trace, and building-specific linear contribution view where available.
- **EDA & Insights:** data coverage, early-weight relationships, target attainment, and limitations.
- **outputs/model_ready/Project_Canary_Model_Ready_Data.xlsx**: outcomes, exact training rows, data dictionary, and daily audit.
- **notebooks/Project_Canary_Harvest_Recovery_Model.ipynb**
- **notebooks/Project_Canary_Day35_Weight_Model.ipynb**
- **models/harvest_recovery_champion.pkl**
- **models/day35_weight_champion.pkl**

Outstanding farm validation

1. Approve environmental thresholds and interventions.
2. Confirm feed-intake units and targets.
3. Define a consistent bodyweight sampling procedure.
4. Confirm the eventual verified harvest-date and ending-population source.
5. Define acceptable forecast-error limits for management use.